a EMber deep, resolution ladder (deepest cell)

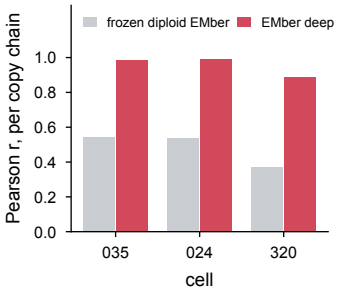
GM12878 scMicro-C cell 035 (GSM5764669), 3.01 M contacts; tag ember-deep-frozen; one RTX 2080 Ti; r: distance r of random particle pairs

final resolution	particles (M) grid / simulated	contacts used (M) (% of input)	models	wall time (min)	seed 1 vs 2, r per copy / whole	vs authors' hickit, r	authors' hickit runs vs each other
100 kb	0.06 / 0.06	1.20 (40 %)	10	1.9	0.98 / 0.99	0.87	0.99
20 kb	0.29 / 0.27	1.15 (38 %)	10	4.2	0.98 / 0.99	0.83	0.99
10 kb	0.58 / 0.51	1.12 (37 %)	10	7.1	0.98 / 0.99	0.81	0.99
5 kb	1.16 / 0.90	1.10 (37 %)	10	11.8	0.98 / 0.99	0.78	0.98
2 kb*	2.90 / 1.57	1.09 (36 %)	10	19.4	0.97 / 0.99	–	–
1 kb*	5.80 / 2.08	1.09 (36 %)	10	22.6	0.97 / 0.99	–	–

* exploratory. r per copy chain: median over the 46 copy chains. Contacts used: committed to a copy pair by the prior or, after re-admission at 200 kb, by the structure; the others are left out.

b Development cells

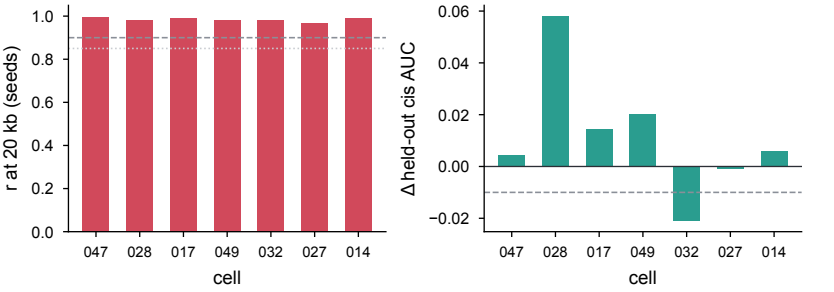
run-to-run r at 20 kb



c Test: 7 held-back cells

C1: r at 20 kb (median ≥ 0.90, each ≥ 0.85); C2: held-out cis AUC − NucDynamics (mean ≥ −0.01)

C1 met: median 0.983, min 0.969. C2 met: mean Δ +0.0117.



Accepted costs (PREREG §11): fewer contacts carry a phase — on the Dip-C development cells the masked ends that receive a phase fall from 0.94 to 0.73 and the correct fraction over all masked ends from 0.83 to 0.71; on scMicro-C only the contacts committed to a copy pair are used (36-45 % of the input on the test cells). The held-out criterion is met on the mean; per cell it varies in both directions (strict positives per cell: 28-50, so per-cell AUCs are noisy). Agreement with the authors' structures is descriptive: they are hickit's own output.